

## 2-2a Thinking, Problem Solving, Judgements, Decision Making

### Concepts & Creativity

How do we organize learned knowledge?

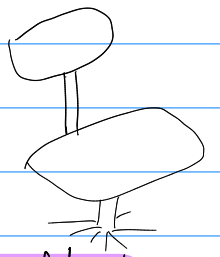
Cognition refers to all mental activities associated w/ thinking, knowing, remembering & communicating info

This incl. meta cognition :: thinking about thinking :  
i.e. thinking when planning & assessing our understanding performance.

Concepts:

We form Concepts :: mental groupings of similar objects, events, ideas or people.

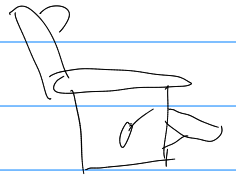
ex. Concept of chair = {office chair, recliner, etc...}



Concepts give us lots of info for low cognitive effort.

Prototypes:

We often form concepts using prototypes :: mental image or best example of a category.



We categorize people into our prototypes

Sometimes, things don't fit nicely to prototypes so we are slow to categorize them

this means that behavior that doesn't fit our discrimination prototypes, ex. female prejudice against females, goes unnoticed.

## Schemas:

Jean Piaget introduced the idea that the maturing brain builds schemas :: concepts or mental molds where we pour our experience.

We assimilate new experiences: we interpret them according to our current schemas.

We accommodate our schemas: adjust them to incl. new info from new exp.

## What is Creativity?

Creativity is the ability to produce novel and valuable ideas.

Creativity is supported by a certain level of aptitude but that's not all

Aptitude tests (ex. SAT) require convergent thinking :: ability to provide single correct answer

Creativity tests require divergent thinking:  
ability to consider many different options & think in novel ways

Functional Fixedness occurs when prior experiences inhibit our ability to find creative solutions

Robert Sternberg believes creativity has 5 components:

Expertise - well developed knowledge. Furnishes the ideas, images & phrases we use as mental building blocks  
More blocks → more novel ways to combine them

Imaginative Thinking Skills - ability to see things in novel ways, recognize patterns & make connections

A venturesome personality - to seek new experiences, tolerate ambiguity & risk, & perseveres in overcoming obstacles.

Intrinsic Motivation - quality of being driven more by interest, satisfaction & challenge than external pressures

Creative Environment - having mentorship, challenge & support from colleagues

## Tips to Improve Creativity

Develop Expertise

Allow time for incubation

Set aside time for mind to roam

Experience other cultures & ways of thinking

## Personal Connection

The ideas of Concept, Prototype & Schema are analogous to computer science concepts.

a Concept is like a class, which is a template that holds data. a Prototype is a default object instance of a class, & a schema can be thought of as a framework, a reusable set of tools & conventions

## MCQs

1. d 2. a 3. a 4. c 5. b 6. b